OIML Certificate

OIML Member State
The Netherlands

Number R49/2013-A-NL1-19.01 revision 7
Project number 4086814

Issuing authority Person responsible:	NMi Certin B.V. M.Ph.D. Schmidt
Applicant and Manufacturer	Euromag International S.r.l. Via Balla, 55 35010 Villafranca Padovana (PD) Italy
Identification of the certified type	An electromagnetic **water meter** Type: MUT1000EL, MUT2200EL and MUT2300 with electronic converter MC406M and MC406AM
Characteristics	See page 2 and further

This OIML Certificate is issued under scheme A.

This Certificate attests the conformity of the above identified type (represented by the sample(s) identified in the OIML Type Evaluation Report) with the requirements of the following Recommendation of the International Organization of Legal Metrology (OIML):

R49-1 (2013) "Water meters intended for the metering of cold potable water and hot water"

Accuracy class 1 and 2

This Certificate relates only to the metrological and technical characteristics of the type of measuring instrument covered by the relevant OIML International Recommendation identified above. This Certificate does not bestow any form of legal international approval.

This certificate and supporting reports comply with the requirements of OIML-CS-PD-07 clause 6.2.

Important note: Apart from the mention of the Certificate's reference number and the name of the OIML Member State in which the Certificate was issued, partial quotation of the Certificate and of the associated OIML Type Evaluation Report(s) is not permitted, although either may be reproduced in full.

Issuing Authority **NMi Certin B.V., OIML Issuing Authority NL1**
8 October 2026

Certification Board

NMi Certin B.V.
Thijsseweg 11
2629 JA Delft
the Netherlands
T +31 88 636 2332
certin@nmi.nl
www.nmi.nl

This document is issued under the provision that no liability is accepted and that the applicant shall indemnify third-party liability.

The notification of NMi Certin B.V. as Issuing Authority can be verified at www.oiml.org

Reproduction of the complete document only is permitted.

This document is digitally signed and sealed. The digital signature can be verified in the blue ribbon at the top of the electronic version of this certificate.

OIML Member State
The Netherlands

Number R49/2013-A-NL1-19.01 revision 7
Project number 4086814

The conformity was established by the results of tests and examinations provided in the associated reports:

- No. NMi-15200444-01 dated 31 March 2016 that includes 39 pages;
- NMi-16200309-01 revision 1 dated 22 March 2018 that includes 29 pages;
- No. NMi-1902198-01 dated 3 December 2018 that includes 16 pages;
- No. NMi-2186686-01 dated 24 May 2019 that includes 43 pages;
- No. NMi-2186686-02 dated 24 May 2019 that includes 47 pages;
- No. NMi-2463352-01 dated 18 January 2021 that includes 27 pages;
- No. NMi-3369067-01 dated 10 May 2022 that includes 13 pages.
- No. NMi-3670720-01 dated 22 September 2023 that includes 12 pages.
- No. NMi-4086814-01 dated 8 October 2026 that includes 21 pages.

Characteristics of the measuring instrument
In Table 1 the general characteristics of the measuring instrument are presented.
The measurement sensor can have the following cylindrical measuring tube:

- Full bore for type MUT1000EL (without flanges) or MUT2200EL (with flanges); or
- Reduced bore for type MUT2300.

Table 2 and 3 gives an overview of the general characteristics of the family of instruments.
The construction of the measuring instrument is recorded in the Documentation folder no. T10713-10.

Table 1 General characteristics

Measuring principle	Electromagnetic
Accuracy class of MUT2200EL	2
Accuracy class of MUT2300	1
Environmental class	M1 / O (installed outdoors)
Electromagnetic environment	E1 for remote version of converter MC406M E2 for compact version of converter MC406M E2 for compact and remote version of converter MC406AM E1 for compact and remote version of converter MC406M/AM with Bluetooth&RS485 and/or 4-20mA output
Temperature range ambient	-25 °C / +55 °C
Water temperature class	T50 (+0,1 °C / +50 °C)
Maximum admissible pressure (MAP)	1,6 MPa (16 bar)
Orientation	All positions (Horizontal, vertical or diagonal)
Flow profile sensitivity class	U0 and D0 (0 x DN upstream and 0 x DN downstream)
Reverse flow	The water meter is designed to measure reverse flow

<table>
<tr><td>Pressure loss class of full-bore type MUT1000EL or MUT2200EL based on documentation 10713/6-01, 10713/6-02 and 10713/12-01.</td><td>Δp 10 for all sizes</td></tr>
<tr><td>Pressure loss class of reduced bore type MUT2300 based on documentation 10713/1-01</td><td>Δp 25 for sizes < DN80
Δp 40 for sizes ≥ DN80</td></tr>
<tr><td>Pressure loss class of reduced bore type MUT2300 based on documentation 10713/4-03, 10713/12-02 and 10713/12-03.</td><td>For sizes < DN250: see below table
For sizes ≥ DN250: Δp 40 (0,040 MPa or 0,40 bar)
<table>
<tr><th>Meter size</th><th>Pressure loss class</th><th>Alternative pressure loss class</th></tr>
<tr><td>DN50</td><td>Δp 25</td><td>Δp 10 for Q3 ≤ 25 m³/h</td></tr>
<tr><td>DN65</td><td>Δp 25</td><td>Δp 10 for Q3 ≤ 40 m³/h</td></tr>
<tr><td>DN80</td><td>Δp 40</td><td>Δp 16 for Q3 ≤ 63 m³/h</td></tr>
<tr><td>DN100</td><td>Δp 40</td><td>Δp 16 for Q3 ≤ 100 m³/h</td></tr>
<tr><td>DN125</td><td>Δp 16</td><td>N/A</td></tr>
<tr><td>DN150</td><td>Δp 40</td><td>Δp 16 for Q3 ≤ 250 m³/h</td></tr>
<tr><td>DN200</td><td>Δp 16</td><td>N/A</td></tr>
</table></td></tr>
<tr><td>Power supply</td><td>Replaceable battery (2,9 – 3,7 V)
DC mains (10 - 28 V) only for MC406AM</td></tr>
<tr><td>Software identification</td><td>Software ‘Bootloader’:
<table>
<tr><th>Software versions</th><th>CRC Checksum</th></tr>
<tr><td>01.00</td><td>63A2EDED</td></tr>
<tr><td>01.01</td><td>67AEA1E4</td></tr>
<tr><td>01.02</td><td>DE7A99AB</td></tr>
<tr><td>01.02</td><td>0CC44DD4</td></tr>
</table></td></tr>
</table>

Software identification	Software 'Legally relevant firmware': **Software versions** / **CRC Checksum** 01.05 / CAA8A4C7 01.15 / 6AA50C55 01.16 / E93E3A1E 01.21 / 79413617 01.23 / E7DD52E4 01.36 / E1A52981 01.38 / C2641A99 01.42 / 2B6CF4C7 01.44 / D867EC79 01.47 / 7144DCB7

Table 2 General characteristics of the family of instruments – Full bore type MUT1000EL or MUT2200EL

Meter size	Ø in- and outlet [mm]	Flow rates [m³/h]				Ratio Q3/Q1
		Minimum Q1	Transitional Q2	Permanent Q3	Overload Q4	
DN32	32	0,125	0,2	25	31,25	200
DN40	40	0,2	0,32	40	50	200
DN50	50	0,315	0,504	63	78,75	200
DN65	65	0,5	0,8	100	125	200
DN80	80	0,8	1,28	160	200	200
DN100	100	1,25	2	250	312,5	200
DN125	125	2	3,2	400	500	200
DN150	150	3,15	5,04	630	787,5	200
DN200	200	5	8	1000	1250	200
DN250	250	8	12,8	1600	2000	200
DN300	300	12,5	20	2500	3125	200
DN350	350	15,625	25	2500	3125	160
DN400	400	20	32	4000	5000	200
DN450	450	31,5	50,4	6300	7875	200

Meter size	Ø in- and outlet [mm]	Flow rates [m³/h]				Ratio Q3/Q1
		Minimum Q1	Transitional Q2	Permanent Q3	Overload Q4	
DN500	500	39,375	63	6300	7875	160
DN600	600	39,375	63	6300	7875	160

Please note that the flow rates Q1, Q2, Q3 and Q4 can be freely chosen as long as:
- Values Q3 and ratio Q3/Q1 are selected from paragraph 4.1 of OIML R49-1: 2013(E);
- Values mentioned for Q1 and Q2 are minimum values and the ratio Q2/Q1 = 1,6;
- Values mentioned for Q3 and Q4 are maximum values and the ratio Q4/Q3 = 1,25;
- The ratio Q3/Q1 is at least 40.

Table 3 General characteristics of the family of instruments - Reduced bore type MUT2300

Meter size	Ø in- and outlet [mm]	Flow rates [m³/h]				Ratio Q3/Q1
		Minimum Q1	Transitional Q2	Permanent Q3	Overload Q4	
DN50	50	0,126	0,203	40	50	315
DN65	65	0,2	0,32	63	78,75	315
DN80	80	0,317	0,5	100	125	315
DN100	100	0,5	0,81	160	200	315
DN125	125	0,5	0,81	160	200	315
DN150	150	1,27	2,03	400	500	315
DN200	200	2	3,2	630	787,5	315
DN250	250	5	8	1000	1250	200
DN300	300	8	12,8	1600	2000	200
DN350	350	10	16	1600	2000	160
DN400	400	12,5	20	2500	3125	200
DN450	450	20	32	4000	5000	200
DN500	500	25	40	4000	5000	160
DN600	600	25	40	4000	5000	160

Please note that the flow rates Q1, Q2, Q3 and Q4 can be freely chosen as long as:
- Values Q3 and ratio Q3/Q1 are selected from paragraph 4.1 of OIML R49-1: 2013(E);
- Values mentioned for Q1 and Q2 are minimum values and the ratio Q2/Q1 = 1,6;
- Values mentioned for Q3 and Q4 are maximum values and the ratio Q4/Q3 = 1,25;
- The ratio Q3/Q1 is at least 40.

OIML Member State
The Netherlands

Table 4 General characteristics of the indicating device - Full bore type MUT1000EL or MUT2200EL

Meter size	Indicating range (minimum value) [m³]	Verification scale interval (minimum resolution) [m³]
DN32	99 999	0,0001
DN40, DN50	99 999	0,001
DN65, DN80, DN100	999 999	0,001
DN125, DN150	999 999	0,01
DN200, DN250, DN300	9 999 999	0,01
DN350, DN400, DN450, DN500, DN600	9 999 999	0,1

Table 5 General characteristics of the indicating device - Reduced bore type MUT2300

Meter size	Indicating range (minimum value) [m³]	Verification scale interval (minimum resolution) [m³]
DN50	99 999	0,0001
DN65	99 999	0,001
DN80, DN100, DN125, DN150	999 999	0,001
DN200	999 999	0,01
DN250, DN300, DN350, DN400	9 999 999	0,01
DN450, DN500, DN600	9 999 999	0,1

OIML Certificate

OIML Member State
The Netherlands

Number R49/2013-A-NL1-19.01 revision 7
Project number 4086814

Certificate history:
This revision replaces the previous version.

Revision	Date	Description of the modification
Initial	24 May 2019	-
1	4 February 2021	Addition of MUT1000EL and full-bore sizes DN32 and DN40 Lowering pressure loss class of MUT2300
2	10 May 2022	Addition of electronic hardware module (Bluetooth&RS485 and/or 4-20mA output). Addition of new software.
3	7 July 2022	Software update
4	14 February 2023	Software update
5	22 September 2023	Update of electronic hardware module. software update
6	10 August 2026	Address change, addition of new sizes, increase of Q3 values, addition of new software and minor hardware change.
7	8 October 2026	Correction to Table 3 – Q2 and Q1 values of the R315 water meters.